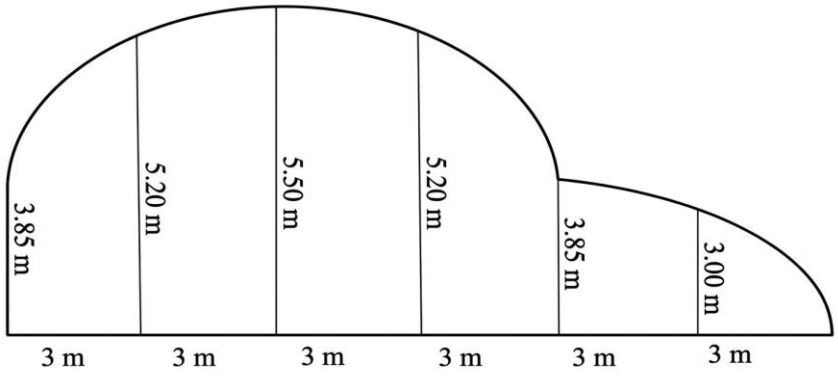
UGANDA MARTYRS SS NAMUGONGO (UMSSN) MATH SEMINAR 2026
PRINCIPAL MATHEMATICS
PAPER 2 ITEMS

CONSTRUCT: NUMEICAL ANALYSIS

Item One

Ivan, a civil engineer from Kampala, has been contracted by the local town council to design and construct a modern, eco-friendly community market in Wakiso District. The project aims to improve local trade while relying entirely on green energy.

To prevent the lower-ground storage units of the busy market from overheating, Ivan designed an underground ventilation and delivery tunnel. Before the excavation team can pour concrete for the walls and ceiling, Ivan needs to estimate the cross-sectional area of this tunnel to order the most suitable quality of reinforcement materials from a supplier.
He measures the height of the tunnel from the ground up to the curved ceiling at equal intervals of 3 $meters$, starting from one baseline edge to the other. His recorded measurements are shown in the diagram below:



Ivan needs to estimate the cross-sectional area of the tunnel to 3 decimal places. If the estimated total area exceeds 75 m^2, high-tensile steel reinforcement will be required.

With the tunnel design finalized, Ivan turns his attention to powering the market. To cut down on electricity bills, the town council decides to install solar panels on the large, flat rectangular roof of the main commercial building. Ivan measures the rooftop dimensions to calculate the area available to be covered by the panels. Due to some water tanks on the roof, a specific section cannot be covered. The company uses the formula $A = w\,(l - c)$ to find the net area to be covered by panels, where:

- $w = 27.56\,m$ is the measured width of the rooftop with percentage error of 4.2%
- $l = 41\,m$ is the measured total length of the rooftop with percentage error of 5%
- $c = 6.25\,m$ is the measured length not to be covered by solar panels with percentage error of 0.8%.

To ensure the procurement of the solar panels stays strictly within the tight council budget, the planning department needs to know the range of possible net area to be covered by solar panels, as well as the percentage error in the area obtained using these measured values. The available budget can only support a net area within the interval 850 m^2 and 1100 m^2 to be covered by the panels.

Task:

Using clear mathematical evidence, help Ivan to determine whether:

a) high-tensile steel reinforcement will be required.

b) the available budget is sufficient.

Item Two

Tom is an operations manager at the NWSC Ggaba Treatment Plant. He is currently resolving three technical challenges that require data-driven mathematical decisions:

A software malfunction has caused a linear shift in chemical dosage readings. A true pump injection rate of $35\ mL/min$ is shown on the screen as $50\ mL/min$, while a true rate of $40\ mL/min$ is displayed as $65\ mL/min$.

- Tom needs to estimate the true injection rate when the screen displays a log of $32\ mL/min$.
- He must also predict the displayed rate when a true injection rate of $38\ mL/min$ is entered.

A new machine will be purchased if the displayed rate for an entered true rate of $38\ mL/min$ exceeds $55\ mL/min$; otherwise, the current machine will be repaired.

Tom also the transmission flow rate pumped up Muyenga reservoir. Water is pumped up to the Muyenga Reservoir at a varying flow rate, $R(t)\ (m^3s^{-1})$, recorded every 4 $hours\ over\ a\ 24-hour$ cycle as shown below:

Time, t (hours)	0	4	8	12	16	20	24
Flow rate, $R(t)\ (m^3s^{-1})$	1.1542	2.4891	3.8215	3.1048	2.9753	1.6340	0.9512

The Muyenga Reservoir has a critical daily storage threshold requirement of $216{,}500.000\ m^3$.If the approximated total volume pumped over the $24-hour$ period is greater than or equal to this threshold, the supply is stable. If it falls below, engineers must activate reserve pumps.

To resolve grid power instability, NWSC installed backup solar panels whose power output, y $(in\ kW)$, is governed by the tilt angle θ (in radians) according to the function:

$$y = 3cos(3\theta)$$

The company intends to lock the panels at a target angle of $\theta = 20°$. Due to tool limitations, the installation team must choose between two tools with the following maximum angle variations:

• Tool A: Angle error of $\pm 0.6°$

• Tool B: Angle error of $\pm 0.2°$

If there's a relative error in the power output y of more than 0.05 of its target value, the system will fail.

Task:

Using clear mathematical evidence, help:

a) Tom to decide whether to buy a new machine or repair the current one.

b) The engineers decide whether to activate the auxiliary pumps.

c) Management choose the correct installation tool to buy for system reliability.

Item Three

An architect is designing a series of underground water culverts. The inner cross-sectional boundary of a culvert is mathematically modeled by the cubic function:

$$\boldsymbol{f(x) = x^3 - 2x^2 - 5x + 8.32}$$

Where $y = 0$ represents the coordinate level at which connecting joint pipes can be physically merged into the primary culvert structure. Preliminary checks show a valid structural junction point exists between $x = 0 \; and \; x = 2$. To get a precise position, the architect needs to design a work flow diagram to compute the root of the function to 3 decimal places using an initial approximation of $x_0 = 1.2$. As a safety precaution, the architect must perform a dry run of the flow diagram to obtain the precise position of the junction point, if this position is below $x = 1.4$, high grade joints ae needed; otherwise medium grade joints should be used.

To evaluate the capacity of water the culvert can manage, the engineering crew needs to approximate the total cross-sectional area bounded under the curve f(x) from $x = 0 \; to \; x = 2$ using 5 equal subdivisions and if this area is below 5.5 $square \; units$, then the culverts needs to be replaced with bigger one.

Task: Basing on Mathematical justification, determine whether:

a) high-grade or medium-grade joints are required based on the precise location of the junction point.
b) the culvert needs to be replaced with a bigger one.

Item Four

A dairy farm in Ntungamo uses a cylindrical milk collection tank with a measured diameter of 3.40m and a depth of $1.20m$, both recorded to 2 decimal places. The C.E.O only considers buying another tank with more accurate measurements if the percentage error in measuring the volume of this tank is more than 10%.

To ensure effective cooling, the refrigerator temperature control system relies on a functional model:

$$T(x) = 2x - 3\cos\left(\frac{x}{2}\right)$$

Where x is the system setting (measured in radians) and the normal positive setting lies between 1 and 2. The storage department performs a test to confirm the normal positive setting' existence between the usual range and the supervisor uses straight line interpolation twice to locate the exact location for refrigerator's normal positive setting and if it is less than 1.5 then the refrigerator is considered safe for preservation.

For quality testing, an automated filling sequence introduces milk into a mixing tank in increasing increments: Instead of adding a constant amount, milk is introduced in increasing increments: 1 $litre$ is added in the first step, 2 $litres$ in the second step, 3 $litres$ in the third step, and continuing progressively up to 10 $litres$ in the 10 th step. The facility needs to design a diagrammatic step-by-step guide for a computer that will automatically accumulate the total volume of milk added across all steps and calculate average of total milk added after all the steps. If the average is at least 5 $litres$ then the mixing tank is certified as stable.

Task:

a) Justify whether the farm's C.E.O should buy another milk tank?
b) Advise the storage department on whether the refrigerator's setting is safe for preservation?
c) Justify whether the milk mixing tank is stable?

Item Five

A fish-trading company in Jinja is preparing a delivery of fresh fish to Kampala. Before approving the operation, the manager wants to assess the refrigeration system, transport arrangements and financial position of the business.

During the journey, the rate at which the refrigerated truck's net cooling effect changes at time t is modelled by $f(t) = t\cos t$ units per unit time, for $0 \leq t \leq \pi$. Technicians want to estimate the total net cooling effect across six equal time intervals. The system is considered satisfactory if the percentage error in the estimated total net cooling effect is not more than 3%.

The company's express bus has stages along the $Jinja - Kampala\ road$ $10\ km\ apart$. The fares from Jinja to the first, second, third and fourth stages are $Shs.1{,}100, Shs.1{,}500, Shs.1{,}850\ and\ Shs.2{,}000$, respectively. Passenger A had paid to travel to the fourth stage but became sick and was left at a health centre $33\ km$ from Jinja. The company policy is to refund a passenger when the value of the unused part of the journey is at least 5% of the fare paid. Another passenger B has only $Shs.1{,}650$ and wants to travel as far as his money permits. The manager will allow him to board only if he can travel at least $25\ km$ from Jinja.

The company makes an annual profit from Tilapia of $Shs.180\ million$ with a margin of error of $\pm 5\%$, and an annual loss from Nile perch of $Shs.100\ million$ with a margin of error of $\pm 3\%$. The annual tax is $Shs.5\ million$. The manager considers the business financially acceptable if the maximum and minimum tax percentages are within 7.5% of the gross income.

Task: Using Mathematical Evidence, help the manager determine whether;

(a) The refrigeration system is satisfactory
(b) The company will refund the passenger A's money and advise if they should allow passenger B to board.
(c) The business is acceptable financially

Item Six

An international school is installing motion sensor alarms along a perimeter path. The target installation zone from the main building is modeled by the function:

$$\boldsymbol{f(x) = 3x^2 + x - 5}$$

Where x represents the distance in hundreds of meters from the school building. Preliminary measurements show that the ideal sensor position is a root of this function lying between $x = 1\ and\ x = 2$. The technician must verify this root lies within the given measurements and use linear interpolation to find a better approximation to 2 decimal places. If this approximation exceeds 150 meters, the installation must be halted for fresh measurements.

During installation, a concrete base mixture is prepared. The recorded mass of cement is $m = 48.50\ kg$ and the volume of water is $v = 120.0\ liters$. The mixture concentration is determined by $c = \frac{m}{v}$. The concrete foundation is considered structurally firm only if the maximum possible percentage error in the calculated concentration is less than 0.7%.

To manage the security project's budget, the administration needs to design a diagrammatic action plan that computes and prints the tax deduction (T) for site workers based on their gross earnings (A) in Uganda Shillings (Ugx). Their computerized payroll algorithm uses the following progressive tax structure:

$\boldsymbol{Amount\ Earned\ A\ (Ugx)}$	$\boldsymbol{Tax\ Paid\ (T, Ugx)}$
$A < 350{,}000$	T = 0
$350{,}000 \leq A \leq 600{,}000$	$T = \frac{3A}{100}$
$600{,}000 \leq A \leq 800{,}000$	$T = \frac{4A}{100}$
$A > 800{,}000$	$T = 80{,}000 + \frac{6}{100} \times (A - 800{,}000)$

Four specific worker salaries to be processed are and their gross earnings are given below:

- $Worker\ 1$: $Ugx\ 250{,}000$
- $Worker\ 2$: $Ugx\ 400{,}000$
- $Worker\ 3$: $Ugx\ 750{,}000$
- $Worker\ 4$: $Ugx\ 1{,}050{,}000$

If the accumulated Total Tax Paid for these four workers is less than $Ugx\ 150{,}000$, the school policy recommends hiring more site workers.

Task:

As a Mathematics student, interpret, show the methodology and make judgements to the above challenges.

Item Seven

Uganda Electricity Transmission Company through Rural Electrification plans to construct a substation in Luwero District to help their society with power stability. The company was allocated an area modelled by the function $y = xe^x$ between the boundaries $x = 0.5\ km\ and\ x = 2.5\ km$ where *x* represents horizontal distance in kilometers and *y* represents vertical distance in kilometers. The Company Engineer plans to estimate the area allocated to the Company to 3 decimal places using 5-ordinates. The site meets the substation standards if the approximated area falls between $19\ km^2\ to\ 20\ km^2$.

Furthermore, management will approve construction only if the percentage error of this estimate compared to the exact value is between $4\%\ and\ 5\%$.

An engineer at the Uganda Electricity Distribution Company (UEDCL) is testing a newly constructed substation grid. During peak hours, the net operational stability factor (y) of the grid fluctuates as a function of the operational voltage (x in Megavolts) according to the simulation model below:

$$y = x - cos(5x)$$

To ensure grid stability, the stability factor must be perfectly balanced $(y = 0)$. Previously collected data suggests that this balance point lies within an interval of $[0, 0.5]\ MV$. The engineer's analysis plan involves first graphically locating the root to obtain an initial voltage approximation, and then applying the Newton-Raphson method to refine this value until it stabilizes with a tolerance error of less than or equal to $0.0001\ MV$. Regulatory standards dictate that this substation grid can only safely distribute power to local households if this stable voltage is equal to or greater than $0.3000\ MV$.

Task:

a) Help the Company Engineer to find out whether:
   i) the estimated area lies in the Substation Standard area.
   ii) they can approve the construction

b) Analyze and confirm whether the substation grid can safely distribute power to local households.

Item Eight

In an End of year offer, a real estate dealer declared a 25% discount on all plots, plus an additional 5% cash payment discount. The dealer would like to design a diagrammatic set of instructions to feed into a computer to compute the final amount paid for each item. A buyer intends to purchase two plots with marked prices sh 12,500,000 and sh 34,000,000 on credit and cash terms respectively. The buyer has a strict total budget of sh 35,000,000 for both properties and will only proceed with the joint purchase if the combined final cost does not exceed this budget

For better drainage, the real estate dealer is to construct a drainage channel alongside the plot of land he's selling. On the master plan, the path of the drainage channel is modeled by the curve: $y = 6.55 - 7x + x^2$, while the straight boundary of the plot lies along the $x-\ axis$. To map the layout, the survey team requires a visual representation of the path by sketching the curve for the path of the drainage channel within the interval $[-4,3]$ to pinpoint an approximate crossover point where the channel intersects the boundary of the plot. If this estimated value is less than 1.0, it indicates a wider encroachment of the drainage channel into the plot and the dealer must give a bigger discount for this specific plot.

The real estate dealer needs to secure a specific section of the drainage channel by installing two culverts at positions $x = 2\ and\ x = 3$. To avoid heavy environmental fines, surface area enclosed between these two culverts, the drainage channel path and the plot boundary must not exceed $1.25\ square\ units$. To verify this compliance, the site engineer intends to estimate the area of this section using 6 equal strips.

Task:
Provide clear Mathematical evidence to help the land buyer, real estate dealer and the engineer make informed decisions.

CONSTRUCT: DATA ANALYSIS AND PROBABILITY

Item One

At a certain Secondary School, the school management is reviewing its student assessment programme. The school wants to use statistical information to make decisions about the time students take to complete an assessment, the gender and subject distribution of students, and the consistency of two academic tests. During a previous assessment exercise, the time X, measured in hours, taken by a randomly selected student to complete a mathematics assessment was found to have the cumulative distribution function:

$$F(x) = \begin{cases} 0 & x \leq 0 \\ \frac{k}{2}x^2; & 0 < x < 2 \\ k(6x - x^2 - 6); & 2 < x < 3 \\ 1 & x \geq 3 \end{cases}$$

The management will retain the current three-hour assessment session if the median time taken by students to complete the assessment is not more than $1.8\ hours$ and the expected assessment time is not more than $1.75\ hours$.

The school is also analyzing the composition of its students according to gender and whether they study science subjects. 55% of the students are male, while 30% of the male students study science subjects. Among the female students, 5% study science subjects. The school will give additional attention to supporting female students in non-science subjects if the probability that a randomly selected student is female given that the student does not study science subjects is at least 0.50.

Finally, six students participated in two academic selection tests, Test X and Test Y. The school recorded the order in which the six students were ranked in each test as shown below.

Student	**1**	**2**	**3**	**4**	**5**	**6**
Test X	E	C	B	F	D	A
Test Y	F	A	D	E	B	C

The school will regard the two tests as sufficiently consistent for use together in assessing students if the rank correlation coefficient between the two tests is at least $0.70\ at\ 5\%$ level of significance.

Task: Using mathematical evidence, help the school make informed decisions.

Item Two

Alex is the newly appointed manager of "The Coffee Haven", a popular coffee shop. But faces a mountain of daily operational problems: delivery trucks are frequently late; customer wait times are unpredictable and marketing budgets are tight. To save the café from losing money, Alex decides to use statistical data to track three major events and make critical business decisions.

The first major event of the day is the arrival time of the morning milk delivery. The delivery truck is scheduled to arrive at $6{:}00\ am$, but due to unpredictable city traffic, its arrival delay follows a rectangular distribution over the interval from $0\ to\ 20\ minutes\ past\ 6{:}00\ am$. The coffee shop officially opens to customers at $6{:}15\ am$. Alex will cancel the current contract if the probability of the truck missing the shop's opening time is greater than 0.20 and he will only sign a contract for the upcoming month if the expected arrival time of the truck is not later than $6{:}08\ am$.

Once the doors open, Alex struggles with customer complaints regarding how long it takes to get coffee. The waiting time, X (in minutes) for a customer to receive their order, can be modelled as a continuous random variable with the following probability density function:

$$f(x) = \begin{cases} \frac{3}{16}x^2 & 0 \le x \le 2 \\ \frac{3}{16}(4-x)^2 & 2 \le x \le 4 \\ 0 & elsewhere \end{cases}$$

Corporate standards require that at least 80% of the customers receive their orders within the interval of $1\ to\ 3$ minutes. If the proportion of customers who receive their orders within this interval is less than $80\ \%$, Alex must issue apology discount coupons. Alex must also schedule a mandatory speed-training workshop for the staff if the standard deviation of the waiting time is greater than $0.5\ minutes$.

To solve the problem of afternoon sales, Alex runs a 7-day social media campaign. Each day, Alex posts a different number of promotional flyers online and tracks the daily revenue to see if more online flyers lead to higher sales. The data collected over the week is shown below:

Flyers posted(x)	5	12	8	15	6	11	9
Daily revenue (y, in thousands of shillings)	400	550	420	600	390	510	480

Alex plans to launch a massive, online campaign next month. He will only approve this budget if there is a strong positive rank correlation greater than 0.85, proving that the social media campaign increases revenue.

Task: Help Alex to decide whether:

a) to cancel the current contract and if he should sign a contract for the upcoming month.

b) to issue apology discount coupons and if he has to schedule speed-training workshop for the staff.

c) he should approve the budget for the online campaign.

Item Three

A group of senior four students from riverine secondary school in Kampala took part in various school activities involving academic assessment, mathematics test, sports and geography field trip. During these activities, they encountered situations that required them to apply mathematical knowledge to analyze information and make appropriate decisions. The following situations arose:

The students sat for a Mathematics test and the results obtained by 100 students were recorded as follows

Marks (%)	$10 - 20$	$-< 25$	$-< 40$	$-< 60$	$-< 70$	$-< 95$	$-< 100$
Number of students	10	26	38	48	68	92	100

The mathematics teacher wishes to submit the marks of the learners to the class teacher only if the number of students that exceed the most common mark is more than 70.

More so, a school has a mathematics contest club and some student's wishes to join the club. As part of the selection process, the club organizes a probability game using a special die labeled $1,\ 2,\ 2, 3, 3, and\ 4$. If the chance of a 2 appearing at least 4 times is less than the chance of a 4 appearing at most than 6 times, the student will join the club.

Later, a group of students went on a geography field trip to Jinja to study the Source of the Nile. Due to road works and heavy traffic along the Kampala–Jinja Highway, the travel delay time X (in hours) experienced by their school bus beyond the expected travel time is modeled as a continuous random variable with the probability density function:

$$F(x) = \begin{cases} 0, & x < 0 \\ ax; & 0 < x < 1 \\ btan^{-1}x\,; & 1 < x < \sqrt{3} \\ 1, & x > \sqrt{3} \end{cases}$$

where $a\ and\ b$ are constants. The geography teacher wishes to decide on the return route back to school. He decided that if the probability that the bus delay time deviates from 1 hour by less than $0.5\ hours$ is greater than the expected delay time, he will use the same route back; otherwise, he will use an alternative route.

Task: Use mathematical calculations to help make informed decisions

Item Four

At Addereth High School in Mukono, the physical education department is preparing students for different sports activities and competitions. The teachers and coaches need to use information about students' physical characteristics and performance to make fair decisions when selecting students for sports teams. The following situations arose during the school's sports selection programme.
They recorded the heights of 60 students participating in sports activities. The table below shows the distribution of their heights.

Height (cm)	$140-$	$150-$	$160-$	$170-$	$180-$	$190-$
Number of students	7	6	15	12	7	13

The teacher establishes a rule that a student will be considered suitable for the basketball team if the difference between the expected height and the median height is at most $2\ cm$.

As part of the school's preparation for the Science Olympiad Team, a student wishes to join the team through a selection activity involving a biased coin. For this coin, getting a head is three times as likely as getting a tail. The student is asked to toss the coin $10\ times$.
The club patron's selection criterion is that the student will qualify if the chance of obtaining at least 6 heads is greater than 0.5.

The school is also selecting one player to represent it in a regional table-tennis competition. John and James are competing for the single position. They take turns playing, with John taking the first turn. The probability that John scores a point on his turn is 0.6, while the probability that James scores a point on his turn is 0.7. The game continues until one player fails to score a point, at which point the other player wins. The coach decides that if John's chance of winning on his third attempt is greater than James's chance of winning on his third attempt, John will qualify for the regional squad.

The school agriculture club is also assessing a large batch of pumpkins harvested from the school farm. The mass X, in kilograms, of a randomly selected pumpkin is modelled as a continuous random variable uniformly distributed between $0.5\ kg\ and\ 2.5\ kg$.
The agriculture teacher decides that if the chance that a pumpkin's mass is less than the expected mass is less than 0.3, the batch will be classified as Grade B; otherwise, it will be classified as Grade A.

Task: Carry out the necessary mathematical computations and make the valid informed decisions.

Item Five

The Kasese District Education and Sports Committee organized an annual secondary-school championship which included athletics, academic activities and a football final. The committee collected information from the different activities to help it make decisions about student performance, examination preparedness, crowd management and the efficiency of organizing future events.

During the athletics competition, ten students participated in the short-put event. The committee recorded each student's height and the greatest distance to which the student threw the shot. The results were:

Student	A	B	C	D	E	F	G	H	I	J
Height $x\ (cm)$	122	124	133	138	144	156	158	161	164	167
Distance $y\ (m)$	41	38	52	56	29	54	59	61	63	67

The committee wanted to establish whether students' height could be useful when identifying students with potential in short-put. The committee will consider height useful as an indicator of throwing performance if the Spearman's rank correlation coefficient is at least 0.70.

At the same championship, Otim, a Senior Four student participating in the academic programme, was preparing for his examinations. Based on his previous performance and teachers' assessments, his chances of passing Physics, Chemistry and Mathematics were estimated to be $0.60, 0.75$ and 0.80, respectively. The school wanted to assess his examination prospects and determine whether he might require additional attention in any of the subjects. The committee will recommend additional Chemistry support if the probability that Otim fails Chemistry, given that he passes at least two subjects, is greater than 0.20.

The championship attracted a large crowd for the final football match. The organizers estimated that 80 % of the spectators supported the home team while 20 % supported the visiting team. For security and crowd-management planning, $2{,}500$ spectators were selected randomly from the crowd. The committee was particularly interested in the possibility of having an unusually large number of visiting-team supporters. The security committee will make additional preparations if the probability of having at least 540 visiting-team supporters is greater than 0.05.

The committee also monitored the time taken to prepare the field before different events. From a sample of 49 preparations, the mean preparation time was 49 $minutes$, with a standard deviation of 2 $minutes$. The committee wanted to ensure that preparation was completed efficiently. The committee will consider the preparation process satisfactory if the 80% confidence interval for the true mean preparation time has an upper limit not exceeding 49.25 $minutes$.

Task: As a student of Mathematics, help the committee make valid informed decisions.

Item Six

A printing company produces examination booklets for schools. The company aims to produce booklets with a target mass of 500 g. Management believes that the production process is satisfactory. However, several schools have complained to the quality control department that many of the booklets they receive appear lighter than expected and are easily torn as students write on them.

To investigate these complaints, the quality control officer randomly selected 20 booklets and recorded their masses, in grams, as shown below:

$490,\ 505,\ 475,\ 520,\ 510,\ 495,\ 485,\ 530,\ 500,\ 515$
$470,\ 490,\ 505,\ 525,\ 480,\ 495,\ 510,\ 500,\ 520,\ 485$

The management has established that the production process will be suspended for a month if the mean mass of the sample exceeds the target mass of 500 g and the standard deviation is greater than 10 g.

To determine the frequency of defective booklets in the company's production, the quality control officer randomly selects 4 booklets from a large production batch. It has been established that any booklet whose mass is less than 480 g is classified as defective.

If the probability of selecting at least 3 defective booklets among the 4 selected booklets is more than 1 %, the occurrence of defective booklets will be considered usual. In that case, the quality control department will direct the treasury department to refund the customers 50% of the payments they made for the booklets already delivered.

One of the customers subsequently filed a complaint about the matter with the Uganda National Bureau of Standards (UNBS). In their defense, the company claimed that the masses of the booklets produced are normally distributed. To verify this claim, UNBS decided to monitor the company's production process by taking a random sample of 36 booklets. If the probability that the sample mean mass is less than $495\ g$ is greater than 5%, UNBS would consider the production process to be in violation of the permitted variation in booklet masses and take the appropriate regulatory action.

Task: As a mathematics student:

a) Advise the quality control officer whether the production process will be suspended.
b) Establish whether the treasury department should prepare for the refund.
c) Establish whether UNBS should take the stated regulatory action.

Item Seven

M & F Manufacturing Company situated in Bukomansimbi town is reviewing its production operations before making decisions about expansion. The management is assessing its work force, financial performance and production quality to determine whether the company is ready for expansion.

The company employs workers whose ages are recorded as follows:

Age (years)	22	23	24	25	26	27
Number of employees	6	8	12	10	9	5

The Management will recruit additional workers if the standard deviation of the employees' ages is greater than $1.5\ years$.

In its sugar-packing section, the company prepares 8,000 packets per week. The table below represents the number of packets sold in randomly selected weeks and their corresponding probabilities.

Number of packets sold	5,000	6,000	7,000	8,000
Probability	0.15	0.30	0.40	0.15

Each packet sold earns UGX 1,500, while the fixed weekly production and packaging cost is UGX 7,000,000. The company will expand its operation if the expected weekly profit from packets sold is at least UGX 2,500,000.

The quality-control department also checks the mass of detergent packets labelled $2\ kg$. The mass of randomly selected packets is normally distributed. 15% of the packets have masses above $2.10\ kg$, while 60% have masses between $1.90\ kg\ and\ 2.10\ kg$. The filling machine will continue to be used only if at least 97% of the packets have masses above $1.80\ kg$.

Task: Using your mathematical knowledge, help the company in making informed decisions.

CONSTRUCT: MECHANICS

Item One

Ms. Suzan is the Chief Operations Engineer at the newly automated export hub in the Namanve Industrial Park, Mukono, which processes Ugandan coffee beans. To maximize output and prevent structural wear, she is conducting a comprehensive safety and efficiency audit across three major distribution sections.

Two robotic vehicles (A and B) transfer crates across a warehouse floor. An overhead sensor tracks their position vectors (in meters) at any time t (in seconds):

• Vehicle A: $r_A = [2\text{t}\,\boldsymbol{i} + 4\text{t}\,\boldsymbol{j}]\,m$
• Vehicle B: $r_B = [8\,\boldsymbol{i} + (8 + 2\text{t})\boldsymbol{j}]\,m$
If a collision is possible, Vehicle A must alter its velocity to a new constant velocity of $V_{new} = (4\boldsymbol{i} + 2\boldsymbol{J})\,ms^{-1}$. For safety compliance, the shortest distance between the two vehicles must always fall within the safety interval $7.0\,m \leq d_{min} \leq 9.0\,m$.

Heavy crates (50 kg) of processed coffee slide from rest down a rough ramp inclined at 30° to the horizontal ($\mu = 0.2$), pulled via a smooth pulley system by a hanging 10 kg counterweight. A safety barrier is placed 4.0 meters down the ramp from the crate's starting position. To avoid structural damage, the crate's speed must not exceed $3.5\,m/s$ upon hitting the barrier

Compressed blocks of waste from coffee process, sing are launched by a mechanical launcher at an angle of 30° above the horizontal with an initial speed of $8.0\,m/s$ towards a furnace incinerator. The incinerator intake sits at a height of $0.8\,m$ above the launch point. To enter safely, the blocks must hit the intake at a horizontal distance between $3.8\,m\ and\ 4.4\,m$ from the launcher and if they land outside this range, the furnace incinerator must be repositioned.

Task:
As an Assistant Safety Consultant, with mathematical evidence assist Ms. Suzan to make valid decisions.

Item Two

A passenger and cargo ferry operates on Lake Victoria between Bukakata and Luuku in Uganda. The vessel Because safe loading and navigation are important in water transport, a team of engineers has been asked to carry out a mathematical assessment before a proposed cargo operation. The assessment involves the stability of a loading platform, the relative motion of two vessels and the effect of water current on the vessel's course.

At the proposed loading facility, engineers use a smooth metallic platform to move cargo into position suitable for loading. To determine an appropriate inclination for the platform, they conduct a model test using a $10\,kg$ mass resting on the platform. One end of an inextensible string is attached to the mass. The string passes up the line of greatest slope of the platform and over a smooth pulley fixed at the top. The other end of the string is attached to a freely suspended $15\,kg$ mass. The suspended mass is initially $4\,m$ above the floor. The system is released from rest. The suspended mass reaches the floor after $1\frac{3}{7}$ seconds, at which point the string goes slack. For the loading platform to be approved, its angle of inclination must be between $25°\ and\ 35°$, inclusive.

During a separate navigation exercise, the engineers model the relative motion of two vessels on Lake Victoria. At the beginning of the trial, ship A is $10\,km$ from ship B. Ship A is travelling at a constant speed of $24\,km\,h^{-1}$ on a bearing of $N40°\,E$. The bearing of A from B at the beginning of the trial is $N30°\,W$. Ship B travels at a constant speed of $22\,km\,h^{-1}$. The engineers want to select a course for B that will result in the smallest possible separation between the two vessels during the trial. To guarantee safety and avoid any possible collision, the course should be between $15^0\ and\ 17^0\ inclusive,$ north east. The separation should not be less than $6.7\,km$, or else adjustments on the course of B should be made.

The engineers then consider another journey in which another vessel D has to travel towards Island Y. The water current along the proposed route has a velocity of $5\ m\ s^{-1}\ on\ a\ bearing\ of\ N30°\ W$. The engineers require the resultant velocity of the vessel D and the water current to be $6\ m\ s^{-1}\ due\ east$. For this exercise, the vessel's velocity must be less than $10\ m\ s^{-1}$ and the direction should be in the interval $S55°E\ to\ S65°E$. If the calculated speed is $10\ m\ s^{-1}$ or greater and the direction is different from the one recommended, engineers must recommend a change in the proposed navigation conditions.

Task.
Help the engineers make the right decisions.

Item Three

During an emergency relief operation following heavy rainfall, a relief organization is preparing to transport essential supplies to a remote area. The team first needs to load a small supply container onto a relief truck. The container weighs 10 N and rests on a rough loading ramp inclined at 30° to the horizontal. To keep the container stationary and on the verge of sliding up the ramp, a worker applies a steady horizontal force. The coefficient of friction between the container and the ramp is 0.5. The worker must apply a horizontal force that lies between $24\ N$ and $30\ N$ to keep the container in the required position.

After the container has been positioned securely on the truck, another piece of equipment is loaded using a pulley system. A load of 4 kg rests on a smooth horizontal platform and is connected by a light, inextensible string passing over a smooth pulley at the edge of the platform to a freely hanging 2 kg load. The team needs to determine the acceleration of the system and the tension in the connecting string. The lifting equipment safety specifications mandate that the loading operations can only proceed if the system's acceleration does not exceed 3.0 m $s^{-2}$and the cable tension stays below 15 N.

Once the supplies have been loaded and secured, two aircraft, P and Q, are assigned to transport them to the affected region. They fly at the same altitude. Aircraft P travels at 300 km h^{-1}on a bearing of 135°, while aircraft Q travels at 350 km h^{-1}on a bearing of 060°. Initially, aircraft P is 10 km directly north of aircraft Q. Civil aviation safety compliance protocols require that the two aircraft must maintain a horizontal separation distance of at least $5\ km$ at all times throughout their flights.

Task:
Analyse the given information and with clear mathematical evidence help in decision making.

Item Four

An aquaculture project on Lake Victoria uses a floating rectangular fish cage, ABCD, managed by a supervisor. The cage measures 12 m along sides AB and CD , and 5 m along sides BC and DA.
Due to strong lake currents and a sudden storm, the supervisor needs to evaluate the structural stability of the cage and the safety of its lifting mechanism to ensure safe operations.
To secure the rectangular cage against currents, six underwater forces act simultaneously along its boundaries and diagonals:

- $80\ N\ along\ side\ AB$
- $60\ N\ along\ side\ BC$

- $40\ N\ along\ side\ CD$
- $20\ N\ along\ side\ DA$
- $130\ N\ along\ diagonal\ AC$
- $65\ N\ along\ diagonal\ BD$

An emergency anchor must be deployed if the magnitude of the net resultant force is $155\ N$ or greater, OR if its direction angle falls outside the range of $45°\ to\ 50°$.

To protect the cage from underwater current damage, a frictionless rope-and-pulley system is installed directly above the water. This frictionless pulley system is designed to raise the cage closer to the surface or lower it deeper by adjusting weights on the opposite end. Currently, the weight of the rectangular cage is $500\ N$ one side of the pulley, counterbalanced by a heavy $60\ kg$ concrete weight block on the other side. Initially, the system is held stationary with both the cage and the weight block suspended exactly $30\ cm$ above the lake bottom. When released from rest, the heavier concrete block falls to the lake bottom. Directly above the rising cage is a rigid steel docking point located exactly $35\ cm$ above the cage's initial starting position. The supervisor needs to determine if the docking point is safe and will not be hit by the cage.

A heavy storm develops, and the supervisor needs to determine the true velocity of the wind to guide nearby vessels:

- The supervisor's motorboat travels at $12\ km/h$ in the direction $S30°W$, observing the wind appearing to blow from $N60°W$.
- A nearby crew ferry travels at $15\ km/h$ in the direction $S80°E$, observing the wind appearing to blow directly from the North.

Maritime guidelines state that fishing vessels must immediately return to the landing site if the true wind speed exceeds $25\ km/h$.

Task:

As a student of Mathematics, assist the supervisor to determine:

a) Whether they need to deploy the emergency anchor based on the net resultant force and angle.
b) Whether the upward-moving cage will collide with and damage the rigid steel docking point.
c) Whether the fishing vessels must immediately return to the landing site or if it is safe to continue their journey.

Item Five

A manager is coordinating an urgent delivery of heavy drilling components to a remote mining site. On his way to the staging ground, the transport vehicle failed, threatening to delay the entire day's operations. He contacted a nearby depot to send a heavy-duty tractor to pull the equipment trailer to a riverside loading dock where it would be loaded onto a cargo vessel to cross a fast-flowing river toward the final installation site.

The heavy-duty tractor of mass $7000\ kg$ was used to tow this equipment trailer of mass $3000\ kg$. On their way, they had to climb an access ramp of $1\ in\ 10$ with the speed at the foot of the incline as $12\ ms^{-1}$ and the tractor's engine exerting a maximum driving force of $22{,}000N$. The ramp is $80\ m$ long and the coefficient of friction between the heavy tires and the ramp surface is 0.06. The tractor's main fuel tank went completely empty just as the system reached the top of the ramp. The tractor needed to then travel on a level road equally rough toward a refueling station to fill the tank. The refueling station was $300\ m$ away from the top of the ramp. The driver needed to know whether to start applying manual brakes so that the heavy convoy doesn't overshoot beyond the

refueling station or to leave it to coast freely if the greatest distance it can travel on a level road is less than the distance to the station.

The convoy finally reached the riverbank, and three ground crew members were called to pull the trailer using steel cables to position it safely onto the cargo vessel using forces of magnitude $600N, 300N\ and\ 400N$ at bearings $030°, 090°\ and\ 120°$. The trailer's structural hitch would be safe from bending or other damages if the net resultant force does not exceed $1010\ N$ and acts at a bearing between $060°\ and\ 075°$.

The manager boarded the cargo vessel that can travel at $4.0\ ms^{-1}$ in still water to cross from Dock A to Dock B directly across the river. The river was flowing at $2.4\ ms^{-1}$ and was $640\ m$ wide, with Dock B situated directly opposite Dock A. The components were eventually delivered and installed at the mining site, but operations started late, affecting the entire production schedule. The manager needs to establish whether the bottleneck was caused by the motion across the river from Dock A to Dock B if the time taken for this crossing is greater than $3\ minutes$, or if the delay was solely due to the mechanical breakdown of the primary vehicle.

Task:

a) Determine whether the heavy convoy safely stops before reaching the refueling station without applying manual brakes.
b) Evaluate whether the trailer's structural hitch remains safe during the alignment process.
c) Establish whether the river crossing caused the operational delay bottleneck based on the $3 - minute$ threshold limit.

Item Six

Moses is a Senior Logistics and Infrastructure Consultant at Port Bell Harbor on Lake Victoria. He is supervising a time-sensitive dispatch operation.

Moses begins by monitoring a heavy-duty delivery truck ($mass\ 1200\ kg$) pulling a cargo supply trailer ($mass\ 300\ kg$) up a steep harbor approach road inclined at a slope of $1\ in\ 10$. The non-gravitational resistance to motion for both the truck and trailer is $0.2\ N/kg$. The truck travels at a uniform constant speed of $1.5\ m/s$ along this final $450\ m$ stretch. To clear the trailer for direct loading, the mechanical coupling must not experience a tension exceeding $400\ N$. Furthermore, the truck enters this incline at exactly $11{:}55\ am$, and the cargo must reach the offloading bay by the strict $12{:}00\ noon$ shipping cutoff.

Upon reaching the harbor's cliffside offloading platform, situated exactly $122.5\ m$ vertically above the water surface, cargo crates are discharged horizontally by an automated hydraulic launcher at an initial speed of $36\ m/s$ into a floating barge grid. For a successful delivery, the crate must land within a designated recovery zone between $175\ m\ and\ 185\ m$ horizontally from the base of the cliff. Additionally, to protect fragile components from structural failure upon impact, the total speed of the crate just before striking the landing surface must not exceed $60\ m/s$.

Moses tracks two incoming transport vessels, Ferry A and Ferry B, navigating the outer harbor channels. The automated tracking system records their constant velocities and positions at different times as:

• Ferry A: Position vector at $11{:}30\ am$ is $\boldsymbol{r}_{\boldsymbol{A}} = (17.5\boldsymbol{i} + 4\boldsymbol{j})\ km$, Velocity vector $\boldsymbol{v}_{\boldsymbol{A}} = (\boldsymbol{i} + 4\boldsymbol{j})\ km/h$
• Ferry B: Position vector $12{:}00\ noon$ is $\boldsymbol{r}_{\boldsymbol{B}} = (4\boldsymbol{i} + 10\boldsymbol{j})\ km$, Velocity vector $\boldsymbol{v}_{\boldsymbol{A}} = (3\boldsymbol{i} + 2\boldsymbol{j})\ km/h$

To prevent harbor accidents, the port authority enforces a strict safety protocol: the two vessels must never cross

within a critical safety clearance distance of 5 km from each other. If they cross this threshold, the port controller must intervene with an emergency rerouting command.

Task:
As the junior engineering consultant attached to this infrastructure project, analyse the given information and with clear mathematical evidence help in decision making.

END